

Decision trees

The fastest way to understand a tree is to train one and visualise it:

```
# Decision Tree: Fit, Inspect, Predict
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree, export_text
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
# Load iris dataset: 150 flowers, 4 features, 3 species
iris = load_iris()
X, y = iris.data, iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

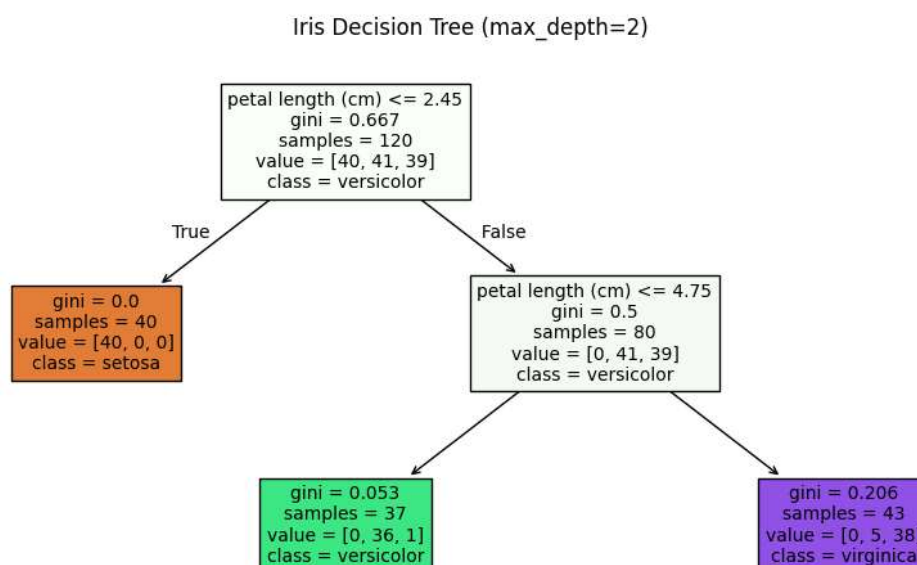
```
# Train a shallow tree (depth=2 so it's tiny and readable)
tree = DecisionTreeClassifier(max_depth=2, random_state=42)
tree.fit(X_train, y_train)
print(f'Train accuracy: {tree.score(X_train, y_train):.2%}')
print(f'Test accuracy: {tree.score(X_test, y_test):.2%}')
```

Train accuracy: 95.00%
Test accuracy: 96.67%

```
# Text view: see the if-else rules as text
print(export_text(tree, feature_names=iris.feature_names))
```

```
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) > 2.45
|   |--- petal length (cm) <= 4.75
|       |--- class: 1
|   |--- petal length (cm) > 4.75
|       |--- class: 2
```

```
# Visual view
plt.figure(figsize=(10, 5))
plot_tree(tree, feature_names=iris.feature_names,
          class_names=iris.target_names,
          filled=True, rounded=False, fontsize=10)
plt.title('Iris Decision Tree (max_depth=2)'); plt.tight_layout(); plt.show()
```



```
# Predict a single new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]] # sepal_len, sepal_wid, petal_len, petal_wid
pred = tree.predict(new_flower)[0]
print(f'Predicted species: {iris.target_names[pred]}')
print(f'Probabilities : {tree.predict_proba(new_flower)[0]}')
```

Predicted species: setosa
Probabilities : [1. 0. 0.]

✓ Gini Impurity

Calculate Gini manually first, then let sklearn demonstrate the same logic:

Gini(node) = 1 - $\sum p_i^2$ where p_i is the proportion of class i

```
# ■ Gini Impurity: Manual + sklearn ■
import numpy as np
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt

# ■ Part 1: Manual Gini calculation ■
def gini_impurity(labels):
    """Calculate Gini impurity for a list of class labels."""
    labels = np.array(labels)
    n = len(labels)
    if n == 0: return 0
    classes = np.unique(labels)
    gini = 1.0
    for c in classes:
        p = np.sum(labels == c) / n # proportion of class c
        gini -= p ** 2
    return gini
```

```
# Test different node compositions
print('Pure node (all same) :', gini_impurity([1,1,1,1])) # 0.00
print('50-50 split :', gini_impurity([0,0,0,1,1,1])) # 0.50
print('70-30 split :', gini_impurity([0,0,0,0,0,0,0,1,1,1])) # 0.42
print('90-10 split :', gini_impurity([0]*9 + [1])) # 0.18
```

```
Pure node (all same) : 0.0
50-50 split : 0.5
70-30 split : 0.42000000000000004
90-10 split : 0.17999999999999994
```

```
# ■ Part 2: Weighted Gini for a split ■
def weighted_gini(left_labels, right_labels):
    """Gini after a split – weighted by group size."""
    n_total = len(left_labels) + len(right_labels)
    w_left = len(left_labels) / n_total
    w_right = len(right_labels) / n_total
    return w_left * gini_impurity(left_labels) + w_right * gini_impurity(right_labels)
```

```
# Two possible splits on a feature – which is better?
# Split A: Glucose <= 120 vs Glucose > 120
left_A = [0,0,0,0,1] # Glucose <= 120: 4 healthy, 1 diabetic
right_A = [0,1,1,1,1] # Glucose > 120: 1 healthy, 4 diabetic
```

```
# Split B: Age <= 30 vs Age > 30
left_B = [0,0,1,1,0] # Age <= 30: mixed
right_B = [0,1,0,1,1] # Age > 30: mixed
```

```
print(f'Weighted Gini – Glucose split: {weighted_gini(left_A, right_A):.3f}')
print(f'Weighted Gini – Age split : {weighted_gini(left_B, right_B):.3f}')
print('Tree picks the split with LOWER weighted Gini → Glucose split is better!')
```

```
Weighted Gini – Glucose split: 0.320
Weighted Gini – Age split : 0.480
Tree picks the split with LOWER weighted Gini → Glucose split is better!
```

```
# ■ Part 3: See sklearn report the Gini per node ■
from sklearn.datasets import make_classification
from sklearn.tree import export_text
X, y = make_classification(n_samples=200, n_features=4, random_state=42)
tree = DecisionTreeClassifier(max_depth=2, random_state=42)
tree.fit(X, y)
# export_text shows 'gini' value at each node
print(export_text(tree, feature_names=['F1', 'F2', 'F3', 'F4']))
# Each node shows: feature threshold | gini | samples | distribution
```

```
|--- F1 <= -0.15
| |--- F3 <= -0.44
| | |--- class: 0
| |--- F3 > -0.44
| | |--- class: 0
```

```

|--- F1 > -0.15
|   |--- F3 <= 0.63
|   |   |--- class: 1
|   |--- F3 > 0.63
|   |   |--- class: 0

```

Plot train vs test accuracy for every depth from 1 to 20. The sweet spot is where they diverge:

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

X, y = make_classification(n_samples=10000, n_features=10, random_state=42)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

depths = range(1, 8)

train_acc, test_acc = [], []

for d in depths:
    tree = DecisionTreeClassifier(max_depth=d, random_state=42)

    tree.fit(X_train, y_train)

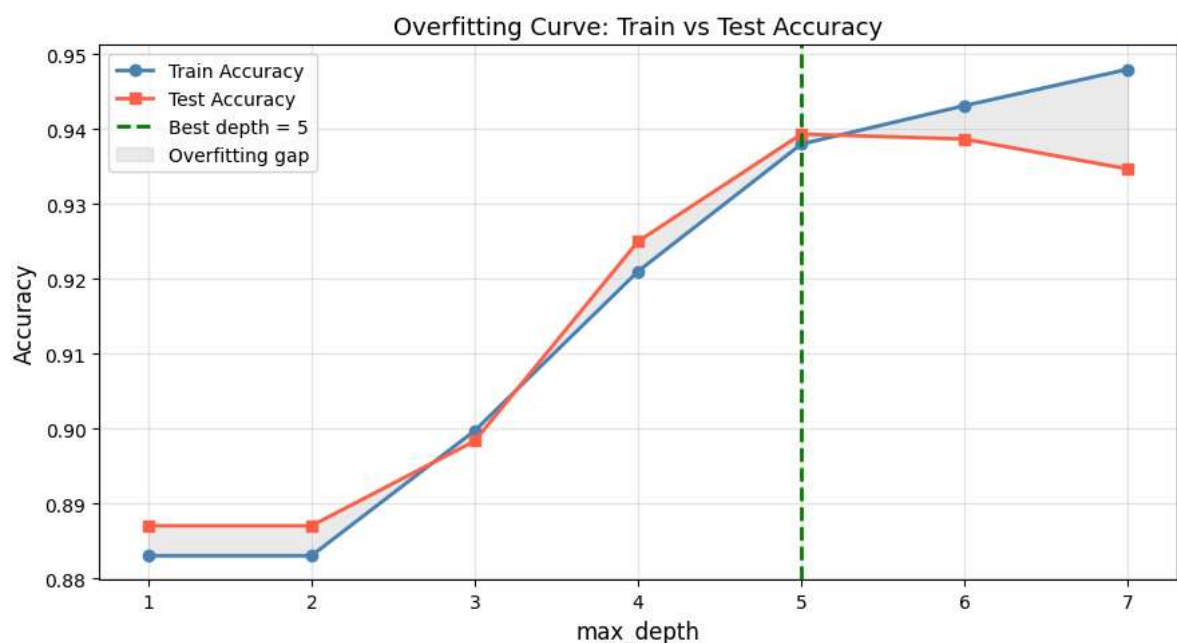
    train_acc.append(accuracy_score(y_train, tree.predict(X_train)))
    test_acc.append(accuracy_score(y_test, tree.predict(X_test)))

best_depth = depths[test_acc.index(max(test_acc))]

plt.figure(figsize=(9, 5))
plt.plot(depths, train_acc, 'o-', color='steelblue', lw=2, label='Train Accuracy')
plt.plot(depths, test_acc, 's-', color='tomato', lw=2, label='Test Accuracy')
plt.axvline(x=best_depth, color='green', linestyle='--', lw=2,
            label=f'Best depth = {best_depth}')
plt.fill_between(depths,
                 train_acc, test_acc,
                 alpha=0.15, color='gray', label='Overfitting gap')
plt.xlabel('max_depth', fontsize=12)
plt.ylabel('Accuracy', fontsize=12)
plt.title('Overfitting Curve: Train vs Test Accuracy', fontsize=13)
plt.legend(); plt.grid(alpha=0.3); plt.tight_layout(); plt.show()

print(f'Best test accuracy {max(test_acc):.2%} at depth={best_depth}')

```



Best test accuracy 93.93% at depth=5

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score

```

```
import numpy as np

configs = [
    {'max_depth': None},
    {'max_depth': 5},
    {'max_depth': 5, 'min_samples_split': 10},
    {'max_depth': 5, 'min_samples_leaf': 5},
    {'max_depth': 5, 'max_features': 'sqrt'},
]
print(f'Config:45s {CV Score (mean±std):20s}')
print('-' * 68)

for cfg in configs:
    tree = DecisionTreeClassifier(**cfg, random_state=42)
    scores = cross_val_score(tree, X, y, cv=5, scoring='accuracy')
    print(f'{str(cfg):45s} {scores.mean():.3f} ± {scores.std():.3f}')
```

Config	CV Score (mean±std)
-----	-----
{'max_depth': None}	0.900 ± 0.004
{'max_depth': 4}	0.918 ± 0.007
{'max_depth': 3, 'min_samples_split': 5}	0.898 ± 0.006
{'max_depth': 2, 'min_samples_leaf': 6}	0.881 ± 0.010
{'max_depth': 1, 'max_features': 'sqrt'}	0.855 ± 0.007